

A counterexample to winner stability in ordinal Chomp

Research report prepared for Vladimir Reshetnikov

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Abstract

Let $\mathcal{S} = \langle 4, 6, 9 \rangle$ be the numerical semigroup ordered by $x \leq_{\mathcal{S}} y$ when $y - x \in \mathcal{S}$. We give a computer-assisted proof that Chomp on $\mathcal{S}$ is a second-player win, whereas Chomp on its ordinal square $\mathcal{S}^2 = \{\omega a + b : a, b \in \mathcal{S}\}$ is a first-player win. An explicit winning first move is $\omega \cdot 4 + 4$. The same move wins on every higher ordinal extension, so $\text{ch}(4, 6, 9) = 2$. This contradicts the winner-stability conjecture in the first version of Rivero Herrera’s paper and answers its current Question 4.6 negatively already for finite natural-number generators.

The infinite-game calculation needs only the first four entries of an option-value spectrum. Its one additional numerical ingredient is that $\text{g}(\text{Ap}(\mathcal{S}, t) \setminus \{0\}) \neq 1$ for every positive $t \in \mathcal{S}$. We certify the stronger inequality $\text{g}(\text{Ap}(\mathcal{S}, t) \setminus \{0\}) \geq 2$ by a deterministic recurrence on 17 gap ideals. Its bounded-Grundy boundary state repeats after 28 steps. Every entry of the finite certificate is also verified by a separate, untruncated finite-poset computation. A general bounded-Grundy eventual-periodicity theorem explains why the certificate proves a universal statement, rather than merely supporting one numerically.

Main conclusion.

$\text{ch}(4, 6, 9) = 2,$	winning first move $\omega \cdot 4 + 4$.
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The proof below includes an exact finite certificate, not an assumed stabilization of a numerical search. The two implementations supplied with this report were executed successfully. The result has not been externally refereed or formalized in a proof assistant.

Keywords. Poset games; numerical semigroups; ordinal powers; Apéry sets; Sprague–Grundy theory; finite-state certificates.

Contents

1	The question and the result	3
1.1	What is being claimed	3
1.2	Proof architecture	3
2	Orders, ordinal extensions, and game conventions	4
2.1	Numerical-semigroup orders	4
2.2	The ordinal extension and its lexicographic description	4
2.3	Poisoned and unpoisoned games	5
3	The game algebra that detects the transition	5
3.1	A three-block detector	6
4	The semigroup and the winning first move	6
4.1	Gaps, symmetry, and the base game	6
4.2	The Apéry diamond	7
4.3	What remains after $\omega \cdot 4 + 4$	7
4.4	An abstract sufficient criterion	8
5	A general finite-state theorem for bounded Grundy values	8
5.1	Gap ideals and boundary positions	9
5.2	The three kinds of moves	9
5.3	Truncation commutes with the necessary mex calculation	10
6	The finite certificate for $\langle 4, 6, 9 \rangle$	11
6.1	The 17 columns	11
6.2	Initial values and the simplified recurrence	11
6.3	The repeated state	12
6.4	What was independently checked	12
7	Concrete replies after the winning move	12
8	Interpretation, scope, and further questions	13
8.1	Why the winner changes	13
8.2	What the certificate does and does not say	13
8.3	Questions left open by this counterexample	14
9	Reproducing the result	14
A	The complete finite-state certificate	14
B	A compact statement of the checking algorithm	16

1 The question and the result

Numerical-semigroup Chomp provides a particularly concrete meeting point of order theory, infinite games, and ordinal arithmetic. A semigroup gives an infinite, locally finite partial order on a subset of the natural numbers. Replacing its elements by lexicographically arranged copies of the same order produces ordinal extensions in which the options can themselves remain infinite. The question is whether that extra order structure can change the winner.

Rivero Herrera’s April 2025 formulation asks whether a second-player win on the first extension necessarily remains a second-player win on all ordinal extensions [3, Conjecture 4.1]. The January 2026 revision asks whether the first winning exponent $\text{ch}(\Gamma)$, when it exists, must have its smallest possible value [4, Question 4.6]. For a finite set Γ of positive natural numbers, that smallest value is 1. The latter question is credited in that revision to Ignacio García-Marco.

Theorem 1.1 (Main result). *Put $S = \langle 4, 6, 9 \rangle$. With the ordinal-extension order defined in section 2, the following hold.*

- (i) *Chomp on $S^1 = S$ is a second-player win.*
- (ii) *Chomp on S^2 is a first-player win. The move $\alpha = \omega \cdot 4 + 4$ leaves a position of Grundy value 0.*
- (iii) *The same move wins on S^σ for every ordinal $\sigma \geq 2$. Consequently $\text{ch}(4, 6, 9) = 2$.*

The first assertion has an elementary proof using the symmetry of this semigroup. The second is the substantive new calculation. It uses a finite certificate for a low-value exclusion, not a simulation of an infinite board. The third follows because the chosen move removes every newly added higher-order block.

1.1 What is being claimed

The assertion is a counterexample within the numerical-semigroup family, not merely within arbitrary pointed partially ordered sets. Winner changes for arbitrary finite pointed posets are already exhibited in [4, Example 3.2]; those examples do not answer the semigroup question. Conversely, this counterexample does not contradict any asserted countable upper bound for a transition exponent: it realizes the very small nontrivial value 2.

The source check for this report found the question still present in the latest arXiv version, dated 5 January 2026, and found no later resolution. That is a statement about the literature located in this investigation, not a claim to have exhaustively searched unpublished work. The mathematical claim is supported by the complete reduction and executable certificate below, independently of its eventual priority assessment.

1.2 Proof architecture

The proof has three logically separate parts. First, an elementary calculation gives the semigroup’s gaps and an Apéry diamond. Second, Sprague–Grundy algebra turns one excluded option value into a winning move in the square. Third, a finite-state calculation establishes that exclusion for every possible first move in the base semigroup.

The infinite conclusion is not obtained by replacing a sufficiently large finite board by an infinite one. No such approximation is used. The only limit argument is the induction that follows an exactly repeated state of a deterministic recurrence.

2 Orders, ordinal extensions, and game conventions

2.1 Numerical-semigroup orders

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. A numerical semigroup is an additive submonoid $\mathcal{S} \subseteq \mathbb{N}$ with finite complement. Its natural semigroup order is

$$x \leq_{\mathcal{S}} y \iff y - x \in \mathcal{S}. \quad (2.1)$$

Here a negative integer is not an element of $\mathcal{S}$. In particular, 0 is the least element, and $x \leq_{\mathcal{S}} y$ implies $x \leq y$ in the ordinary order. The order need not be linear: in our example, 6 and 9 are incomparable because $9 - 6 = 3 \notin \mathcal{S}$.

For positive $t \in \mathcal{S}$, define

$$\text{Ap}(\mathcal{S}, t) = \mathcal{S} \setminus (t + \mathcal{S}). \quad (2.2)$$

This is exactly the surviving board after selecting t . If F is the largest gap of $\mathcal{S}$, every member of $\text{Ap}(\mathcal{S}, t)$ is at most $t + F$, so the board is finite. We write $\text{Ap}(\mathcal{S}, 0) = \emptyset$ when discussing the ordinary, unpoisoned game.

The study of this game, including the symmetric-semigroup second-player result and the finite-boundary representation used later, originates in García-Marco and Knauer [1]. We prove the particular facts needed here rather than treating their general theorems as black boxes.

2.2 The ordinal extension and its lexicographic description

For an ordinal σ , let $\mathcal{S}^\sigma$ consist of 0 and the ordinals whose Cantor normal forms are

$$\omega^{\beta_0} s_0 + \dots + \omega^{\beta_{r-1}} s_{r-1}, \quad \sigma > \beta_0 > \dots > \beta_{r-1}, \quad s_i \in \mathcal{S} \setminus \{0\}. \quad (2.3)$$

Thus $\mathcal{S}^0 = \{0\}$ and

$$\mathcal{S}^1 = \mathcal{S}, \quad \mathcal{S}^2 = \{\omega a + b : a, b \in \mathcal{S}\}.$$

For finite natural-number generators this is the coefficientwise description of the naturally generated ordinal submonoids used in [3, 4].

The partial order is defined using *ordinary ordinal subtraction*:

$$\alpha \leq_{\mathcal{S}^\sigma} \beta \iff \alpha \leq \beta \text{ and } \beta - \alpha \in \mathcal{S}^\sigma, \quad (2.4)$$

where $\beta - \alpha$ is the unique γ with $\alpha + \gamma = \beta$. It is not coefficientwise subtraction under the natural sum.

For the square, equation (2.4) becomes the strict lexicographic rule

$$(a, b) \leq (c, d) \iff a <_{\mathcal{S}} c \text{ or } (a = c \text{ and } b \leq_{\mathcal{S}} d), \quad (2.5)$$

under the identification $(a, b) \leftrightarrow \omega a + b$. Indeed, if $a < c$ then

$$(\omega c + d) - (\omega a + b) = \omega(c - a) + d;$$

if $a = c$ and $b \leq d$, the difference is $d - b$. This also explains why we must write $\omega \cdot 4 + 4$, not $4 \cdot \omega + 4$: these are different ordinals.

2.3 Poisoned and unpoisoned games

For a poset with minimum 0, Chomp makes 0 a poisoned move. Equivalently, we delete 0 in advance and play the normal poset game: a move removes its chosen element and every greater element; a player with no legal move loses. This convention makes the game an ordinary impartial normal-play game.

We distinguish two games on the same semigroup:

$$G = \text{the normal poset game on all of } \mathcal{S}, \quad H = \text{the normal poset game on } \mathcal{S} \setminus \{0\}.$$

Thus H is semigroup Chomp, while G has a legal move at 0. For a finite surviving board D containing 0, its poisoned version is $D \setminus \{0\}$ and its unpoisoned version is D itself.

Every play of either game on $\mathcal{S}$ terminates: its first move either empties the board or leaves a finite Apéry set. After our prescribed move in the square, only finitely many such blocks remain. Once a block is touched it becomes finite, so every continuation also terminates. Thus all the Grundy values used in the main argument are well defined, even though some games have infinitely many options.

3 The game algebra that detects the transition

For a terminating impartial game X , let

$$\mathcal{O}(X) = \{g(X') : X \longrightarrow X'\}, \quad g(X) = \text{mex } \mathcal{O}(X).$$

The mex is taken among ordinals if necessary. A position is a second-player win precisely when its Grundy value is 0. All option values of G are finite, because each of its options is a finite poset.

Write $X \boxplus Y$ for the disjunctive sum. For finite Grundy values,

$$g(X \boxplus Y) = g(X) \oplus_2 g(Y),$$

where $\oplus_2$ denotes binary xor, not Hessenberg's natural sum of ordinals. In the *ordinal sum of games* $X : Y$, a move in X discards Y , whereas a move in Y leaves X untouched. In poset terms, every element of the lower game X lies below every element of the upper game Y .

Lemma 3.1 (Excluded-ordinal formula). *For a set A of ordinals, let $\text{ex}_\eta(A)$ be the η -th ordinal not in A , counting from 0. Then*

$$g(X : Y) = \text{ex}_{g(Y)}(\mathcal{O}(X)). \quad (3.1)$$

In particular, if $g(X) = g(Y) = 0$, then $g(X : Y) = 0$.

Proof. This is the ordinal-sum formula recorded in [2, Theorem 2.6]; the short proof fixes the convention. Put $A = \mathcal{O}(X)$ and induct on the game tree of Y . Moves in X supply exactly the values in A . By induction, moves in Y supply $\text{ex}_b(A)$ for $b \in \mathcal{O}(Y)$. Every ordinal outside A occurs uniquely as some $\text{ex}_b(A)$, and these ordinals increase with b . The first index missing from $\mathcal{O}(Y)$ is $g(Y)$, so the mex of the combined option set is $\text{ex}_{g(Y)}(A)$. The final assertion uses $\text{ex}_0(A) = \text{mex } A = g(X)$. $\square$

Corollary 3.2 (Adding a legal minimum). *If D is a finite poset with least element 0, then*

$$g(D) = 1 + g(D \setminus \{0\}). \quad (3.2)$$

Proof. Write $D = \mathbf{1} : (D \setminus \{0\})$. Since $\mathcal{O}(\mathbf{1}) = \{0\}$, the n -th excluded ordinal is $n + 1$ for finite n . $\square$

3.1 A three-block detector

Lemma 3.3 (The low-value detector). *Suppose $g(G) = 1$, all values in $B = O(G)$ are finite, and*

$$B \cap \{0, 1, 2, 3\} = \{0, 3\}. \quad (3.3)$$

Then

$$g((G \boxplus G) : G) = 3. \quad (3.4)$$

Proof. For $L = G \boxplus G$, we have $g(L) = 1 \oplus 1 = 0$ and

$$C := O(L) = \{b \oplus 1 : b \in B\}.$$

The assumed spectrum gives

$$1, 2 \in C, \quad 0, 3 \notin C.$$

Indeed, these four assertions correspond respectively to $0, 3 \in B$ and $1, 2 \notin B$. All further finite option values are irrelevant to the first two excluded ordinals. Therefore $\text{ex}_0(C) = 0$ and $\text{ex}_1(C) = 3$. Apply theorem 3.1 with upper-game value $g(G) = 1$. $\square$

Remark 3.4 (Why a single nimber is not enough). Replacing *every* occurrence of G by a singleton would be wrong. A singleton also has Grundy value 1, but

$$g((1 \boxplus 1) : 1) = 2,$$

not 3. The lower components contribute their option spectrum, not just their mex. The distinction is systematic in the ordered-join substitution theory of [2]. In this proof the upper component can be summarized by its Grundy value; the lower components cannot.

4 The semigroup and the winning first move

4.1 Gaps, symmetry, and the base game

Lemma 4.1. *For $S = \langle 4, 6, 9 \rangle$,*

$$\mathbb{N} \setminus S = \{1, 2, 3, 5, 7, 11\} =: E. \quad (4.1)$$

Its conductor is 12, its largest gap is $F = 11$, and for $0 \leq n \leq 11$,

$$n \in S \iff 11 - n \notin S. \quad (4.2)$$

Proof. Below 12, the semigroup elements are exactly 0, 4, 6, 8, 9, 10. Also $12 = 4 + 4 + 4$, $13 = 4 + 9$, $14 = 4 + 4 + 6$, and $15 = 6 + 9$. Adding 4 to these four consecutive elements gives every larger integer. This proves equation (4.1); the symmetry assertion is an inspection of the six complementary pairs summing to 11. $\square$

Lemma 4.2. *For every positive $t \in S$, $\text{Ap}(S, t)$ has a greatest element $t + 11$. Consequently*

$$g(H) = 0, \quad g(G) = 1. \quad (4.3)$$

Proof. The integer $M = t + 11$ lies in $\mathcal{S}$, but $M - t = 11$ does not, so M belongs to the Apéry set. Let z be any other member. If $z < t$, then $M - z > 11$, hence $M - z \in \mathcal{S}$. If $z \geq t$, the difference $z - t$ is a gap, and symmetry gives $M - z = 11 - (z - t) \in \mathcal{S}$. Thus $z \leq_S M$.

A nonempty finite normal poset game with a greatest element cannot have Grundy value 0. Otherwise deleting the greatest element would leave a winning position. A winning move from that position could have been played directly in the original position, because it also deletes the greatest element, a contradiction. Apply this argument to $\text{Ap}(\mathcal{S}, t) \setminus \{0\}$, whose greatest element is still M . Every option of H is therefore nonzero, so $g(H) = 0$. Finally, the 0 move in G leaves the empty game, while each positive move leaves a finite board of value at least 2 by theorem 3.2. Its mex is 1. $\square$

This is a direct proof of the symmetric-semigroup phenomenon in this particular case. It does *not* establish the additional exclusion of value 1 from the poisoned Apéry boards. That stronger fact is precisely the finite-certificate lemma proved in section 6.

4.2 The Apéry diamond

Since 4 is the least positive semigroup element, it is an atom in the semigroup order. Directly from equation (4.1),

$$D := \text{Ap}(\mathcal{S}, 4) = \{0, 6, 9, 15\}. \quad (4.4)$$

Its only strict cover relations are $0 < 6 < 15$ and $0 < 9 < 15$. Let D denote the unpoisoned normal game on this four-point diamond. Its options have Grundy values

Move	Remaining value	Remaining shape
0	0	Empty
6	2	The chain $0 < 9$
9	2	The chain $0 < 6$
15	1	A minimum below two incomparable points

Hence

$$g(D) = 3, \quad g(D \setminus \{0\}) = 2. \quad (4.5)$$

4.3 What remains after $\omega \cdot 4 + 4$

Select $(4, 4)$ in the lexicographic square. The surviving outer coefficients are 0, 4, 6, 9, 15. At outer coefficient 4, only the inner diamond D remains. At 0, 6, 9, 15, the whole inner semigroup remains. Every other outer coefficient is removed because it lies strictly above 4 in the semigroup order.

Among the surviving positive outer coefficients, 4 is incomparable with 6, 9, 15: their positive differences from 4 are respectively the gaps 2, 5, 11. Meanwhile 6 and 9 are incomparable and both lie below 15. The global poisoned point is $(0, 0)$, so the outer-0 block is H , not G . Every element of that block lies below all other surviving blocks. Thus the residual game is exactly

$$R = H : (D \boxplus ((G \boxplus G) : G)). \quad (4.6)$$

Proof of theorem 1.1, using the certified lemma. The base-game statement is theorem 4.2. Theorem 6.1 below proves that each positive option of G has value at least 3. The zero move has value 0, and the move 4 has value 3 by equation (4.5). Therefore

$$O(G) \cap \{0, 1, 2, 3\} = \{0, 3\}.$$

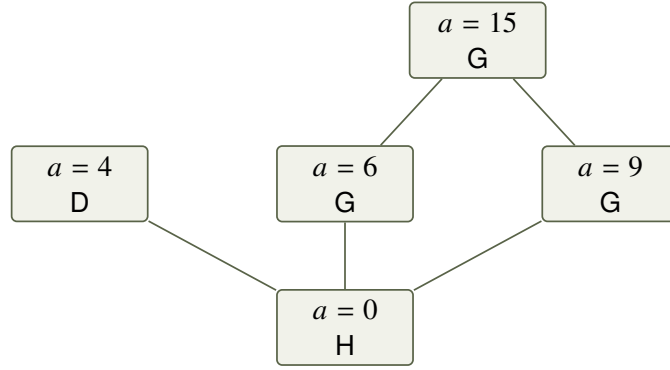


Figure 1: The exact residual block order. Each edge means that every element of the lower block lies below every element of the upper block. The poisoned point is removed only from the $a = 0$ block.

By theorem 3.3, the three-block game $K = (G \boxplus G) : G$ has value 3. Consequently

$$g(D \boxplus K) = 3 \oplus_2 3 = 0.$$

Since $g(H) = 0$, theorem 3.1 and equation (4.6) give $g(R) = 0$. Thus $(4, 4)$ is a winning first move.

Now let $\sigma > 2$. Every new ordinal $\beta \in \mathcal{S}^\sigma \setminus \mathcal{S}^2$ has a leading exponent at least 2. Hence $(\omega \cdot 4 + 4) + \beta = \beta$, so $\beta - (\omega \cdot 4 + 4) = \beta \in \mathcal{S}^\sigma$. The move removes every such β , leaving exactly the same residual game as in the square. Therefore the first winning exponent is 2. $\square$

4.4 An abstract sufficient criterion

The argument isolates a useful pattern that is not specific to semigroups.

Proposition 4.3 (Diamond-atom criterion). *Let P be a pointed poset whose normal and poisoned games terminate, and write G for its normal game. Suppose that:*

- (a) $g(P \setminus \{0\}) = 0$ and $g(G) = 1$;
- (b) P has an atom a for which $P \setminus \uparrow a$ is the four-point diamond;
- (c) $2 \notin O(G)$, and all values in $O(G)$ are finite.

Then (a, a) is a winning first move in poisoned lexicographic Chomp on P^2 .

Proof. The minimum move gives $0 \in O(G)$, the value of G excludes 1, and the diamond option supplies 3. The atom a is incomparable with every positive point of $P \setminus \uparrow a$, since it is neither above a positive predecessor nor below a surviving point. The residual decomposition is therefore exactly equation (4.6), with P in place of $\mathcal{S}$. The detector calculation applies unchanged. $\square$

5 A general finite-state theorem for bounded Grundy values

The remaining task is universal: exclude a value for *every* positive first move, not only for a long initial interval. We now derive the finite-state mechanism that makes this possible.

5.1 Gap ideals and boundary positions

Let $\mathcal{S} \neq \mathbb{N}$ be any numerical semigroup. Put

$$E = \mathbb{N} \setminus \mathcal{S}, \quad F = \max E, \quad c = F + 1.$$

Order the finite set of gaps by $u \preceq v$ if $v - u \in \mathcal{S}$. Let $I(E)$ be the set of its order ideals and let $r = |I(E)|$. For $x \geq c$ and $C \in I(E)$, define

$$D_{x,C} = (\mathcal{S} \cap [0, x)) \cup (x + C), \quad P_{x,C} = D_{x,C} \setminus \{0\}. \quad (5.1)$$

The first missing semigroup element of $D_{x,C}$ is x .

Lemma 5.1. *The sets $D_{x,C}$ are precisely the finite downsets whose first missing semigroup element is an integer $x \geq c$. Moreover,*

$$D_{x,E} = \text{Ap}(\mathcal{S}, x). \quad (5.2)$$

Proof. If D is a downset with first missing element x , every element of $\mathcal{S}$ below x belongs to D . A surviving $x + g \geq x$ cannot have $g \in \mathcal{S}$, since then $x \leq_{\mathcal{S}} x + g$. Thus its offset lies in E . Because $x \geq c$, every $x + h$ with $h \in E$ belongs to $\mathcal{S}$; down-closure then makes the set of surviving offsets an ideal in the gap order.

Conversely, suppose C is a gap ideal and $z = x + g \in x + C$. A predecessor $p \geq x$ has the form $x + h$ with $g - h \in \mathcal{S}$. The offset h cannot lie in $\mathcal{S}$, since that would imply $g \in \mathcal{S}$. Hence $h \in E$, and the ideal property puts h in C . Predecessors below x are already in the prefix. This proves down-closure. Finally, after a move at x exactly the gap offsets survive, giving equation (5.2). $\square$

This boundary description refines the finite-gap encoding in [1, Section 6]: after the conductor, the admissible tails are exactly gap ideals, not arbitrary subsets of the gaps.

5.2 The three kinds of moves

Fix $x \geq c + F$ and $C \in I(E)$. For $1 \leq d \leq F$, put

$$T_d(C) = \{g \in E : g < d \text{ or } g - d \in C\}. \quad (5.3)$$

For $a \in C$, put

$$R_a(C) = \{g \in C : g - a \notin \mathcal{S}\}. \quad (5.4)$$

A negative difference is again treated as not in $\mathcal{S}$. All these sets are gap ideals.

A legal move y in $P_{x,C}$ falls into one of three classes.

A move far inside the prefix: $0 < y < x - F$. The residual is $\text{Ap}(\mathcal{S}, y) \setminus \{0\}$. Every surviving point of that Apéry set lies at most at $y + F < x$, so none is affected by the old tail.

A move near the boundary: $y = x - d$, $1 \leq d \leq F$. Here $y \geq c$, so it is always a semigroup element. A potential surviving point is $y + g$ with $g \in E$. It was present before the move precisely when $g < d$ or $g - d \in C$. Thus the residual is $P_{x-d, T_d(C)}$.

A move in the tail: $y = x + a$, $a \in C$. The whole prefix stays. Among the old tail offsets, exactly $R_a(C)$ stays. The residual is $P_{x, R_a(C)}$. This is a strictly smaller tail, so such states can be processed in increasing order of $|C|$.

These are identities of actual finite posets. They do not depend on any assumptions about which player wins.

5.3 Truncation commutes with the necessary mex calculation

For an integer $k \geq 1$, let $\tau_k(n) = \min(k, n)$. Its value k means “at least k ,” not necessarily the exact value k . For any finite set of nonnegative integers A ,

$$\tau_k(\text{mex } A) = \tau_k(\text{mex}\{\tau_k(a) : a \in A\}). \quad (5.5)$$

Indeed, the answer below k depends only on which of $0, \dots, k-1$ occur. If all occur, both sides are k .

Define

$$q_x(C) = \tau_k(g(P_{x,C}))$$

and record the low values of all far-prefix options by

$$B_x = \{j < k : \text{some } 0 < y < x - F, y \in \mathcal{S}, g(\text{Ap}(\mathcal{S}, y) \setminus \{0\}) = j\}.$$

The move classification gives the exact recurrence

$$q_x(C) = \tau_k \text{mex} (B_x \cup \{q_{x-d}(T_d(C)) : 1 \leq d \leq F\} \cup \{q_x(R_a(C)) : a \in C\}). \quad (5.6)$$

The prefix set updates by

$$B_{x+1} = B_x \cup (\{q_{x-F}(E)\} \cap \{0, \dots, k-1\}). \quad (5.7)$$

The equality uses $x - F \geq c$, so equation (5.2) applies.

Theorem 5.2 (Bounded-Grundy eventual periodicity). *For each numerical semigroup $\mathcal{S} \neq \mathbb{N}$ and each fixed $k \geq 1$, the vector*

$$(\min(k, g(P_{x,C})))_{C \in I(E)}$$

is eventually periodic in x . The determining boundary state has at most

$$N_k = 2^k (k+1)^{Fr} \quad (5.8)$$

possible values. For every prescribed finite nimber j , it is decidable whether some poisoned Apéry board has Grundy value j .

Proof. Before computing row $x \geq c + F$, retain B_x and the previous F rows,

$$U_x = (B_x, (q_{x-F}(C))_C, \dots, (q_{x-1}(C))_C).$$

There are at most 2^k prefix sets and $(k+1)^{Fr}$ row windows. Equations (5.6) and (5.7), with the tails processed in increasing cardinality, determine U_{x+1} from U_x with no remaining dependence on the absolute value of x . A repeated state therefore yields an exactly periodic future. A repetition occurs among any $N_k + 1$ consecutive states.

The finitely many initial rows $c \leq x < c + F$ and initial prefix options are ordinary finite-poset calculations. For a prescribed j , take $k = j + 1$, compute until the first repeated state, and inspect the full-tail coordinate and the exceptional first moves below c . No new value can first occur after the cycle has closed. $\square$

Remark 5.3. The theorem concerns every *fixed bounded range* of Grundy values. It does not assert eventual periodicity of the full, unbounded integer Grundy sequence. The separate cutoffs may have different preperiods and periods. When $\mathcal{S} = \mathbb{N}$, the positions are chains and the bounded statement is immediate without a gap-state construction.

6 The finite certificate for $\langle 4, 6, 9 \rangle$

Lemma 6.1 (Certified low-value exclusion). *For every positive $t \in \langle 4, 6, 9 \rangle$,*

$$g(\text{Ap}(\mathcal{S}, t) \setminus \{0\}) \geq 2. \quad (6.1)$$

Equivalently, every positive option of the unpoisoned game $\mathbf{G}$ has Grundy value at least 3.

We give the precise certificate and its verification argument. It is useful to keep the labels completely explicit:

$0 = \text{exactly } 0,$	$1 = \text{exactly } 1,$	$2 = \text{at least } 2.$
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All rows below concern *poisoned* finite boards; the unpoisoned values are obtained by adding 1.

6.1 The 17 columns

For $E = \{1, 2, 3, 5, 7, 11\}$, the gap-order cover relations are

$$1 \prec 5, \quad 1 \prec 7, \quad 3 \prec 7, \quad 2 \prec 11, \quad 5 \prec 11, \quad 7 \prec 11.$$

Its order ideals, in the exact order used by the certificate, are listed in table 1. In particular, $C_{16} = E$.

i	C_i	i	C_i
0	$\emptyset$	9	$\{1, 2, 5\}$
1	$\{1\}$	10	$\{1, 3, 5\}$
2	$\{2\}$	11	$\{1, 3, 7\}$
3	$\{3\}$	12	$\{1, 2, 3, 5\}$
4	$\{1, 2\}$	13	$\{1, 2, 3, 7\}$
5	$\{1, 3\}$	14	$\{1, 3, 5, 7\}$
6	$\{1, 5\}$	15	$\{1, 2, 3, 5, 7\}$
7	$\{2, 3\}$	16	$\{1, 2, 3, 5, 7, 11\}$
8	$\{1, 2, 3\}$		

Table 1: The 17 admissible gap tails, sorted first by cardinality and then lexicographically.

6.2 Initial values and the simplified recurrence

There are five positive semigroup elements below 12. Their exact poisoned Apéry values are

t	4	6	8	9	10
$g(\text{Ap}(\mathcal{S}, t) \setminus \{0\})$	2	4	3	6	5.

(6.2)

Thus the initial far-prefix low-value set B_{23} is empty for cutoff $k = 2$. The first eleven rows $12 \leq x \leq 22$ are computed directly from the finite boards P_{x, C_i} . The largest integer appearing in these initial boards is 33.

Whenever all earlier full-tail labels are 2, the prefix update keeps $B_x = \emptyset$. Therefore the specialized recurrence is simply

$$q_x(C) = \text{mex}_2(\{q_{x-d}(T_d(C)) : 1 \leq d \leq 11\} \cup \{q_x(R_a(C)) : a \in C\}), \quad x \geq 23, \quad (6.3)$$

where $\text{mex}_2 A = \min(2, \text{mex } A)$. This simplification is *checked inductively*; it is not a prior assumption of the conclusion. Every computed full-tail coordinate must remain 2, and the same condition must hold throughout the certified cycle.

6.3 The repeated state

Write the 17-entry row as the word

$$w_x = q_x(C_0)q_x(C_1) \cdots q_x(C_{16}) \in \{0, 1, 2\}^{17}.$$

The complete certificate consists of the 253 words $w_{12}, \dots, w_{264}$ printed in section A. It has the following three properties:

$$\begin{aligned} q_x(C_{16}) &= 2 & (12 \leq x \leq 264), \\ (w_{226}, \dots, w_{236}) &= (w_{254}, \dots, w_{264}), \\ w_x \text{ satisfies equation (6.3)} & & (23 \leq x \leq 264). \end{aligned} \tag{6.4}$$

The equality compares exactly $11 \times 17 = 187$ labels. It says that the complete boundary state before row 237 equals the complete boundary state before row 265. The period is $265 - 237 = 28$.

Proof of theorem 6.1. The finite recursion verifies equation (6.2), the eleven initial rows, and equation (6.4). Each initial full-tail label is 2, so $B_{23} = \emptyset$. Inductively, the specialized recurrence is the general recurrence and preserves this empty prefix set through row 264. The two complete boundary states now coincide. Determinism implies that every subsequent state follows the same cycle, and each full-tail label in that cycle is again 2. Thus $q_t(E) = 2$ for every $t \geq 12$. Together with the five exceptional first moves below 12 and equation (5.2), this proves equation (6.1). The unpoisoned reformulation follows from theorem 3.2. $\square$

The row sequence is periodic from 226 with period 28. The calculation also checks $w_{225} \neq w_{253}$, so 226 is the earliest onset for this particular period. No assertion about the least possible period is needed.

6.4 What was independently checked

The supplied generator uses sets of surviving integers and truncated recursion. The independent verifier instead rebuilds semigroup membership from the generators and uses bit masks and *untruncated* mex values. It directly solves all $253 \times 17 = 4\,301$ finite boards represented in the certificate, not merely the eleven initial rows. Its cache contains 4 346 finite states in total. The largest board element used in these checks is $264 + 11 = 275$.

The verifier additionally checks 233 literal residual-poset identities, replays the bounded recurrence, checks the 187-entry repeated boundary state, and evaluates the diamond and a finite analogue of the three-block detector. Six regression tests pass, including deliberate certificate corruption, the capped-mex identity, and the boundary recurrence for eight semigroups at three cutoffs. The exact verification output and full integer Grundy tables are included in the archive.

These are separate implementations of the finite calculation, not two independent human proofs. The trust boundary is explicit: ordinary mathematical reasoning plus finite exact integer computations under Python. No floating-point arithmetic, random search, external solver, or conjectured period is involved. No formal proof-assistant verification is claimed.

7 Concrete replies after the winning move

The first move is explicit, and the proof also gives concrete responses to every possible second move. This is useful both as an interpretation of the Grundy calculation and as an additional check on the block decomposition. Let Alice first play $(4, 4)$. Bob now moves in one of the five blocks in figure 1.

Bob moves in the bottom block $a = 0$. Every upper block disappears. The residual is a finite poisoned Apéry board of value at least 2. Alice computes an ordinary finite-poset move to value 0.

Bob moves in the diamond block $a = 4$. Let $d \in \{0, 1, 2\}$ be its remaining normal Grundy value. Alice changes the three-block component K to the same value d :

d	Alice's move	New three-block value
0	$(15, 0)$	$g(G \boxplus G) = 0$
1	$(6, 0)$	$g(G) = 1$
2	$(6, 4)$	$g(D \boxplus G) = 3 \oplus_2 1 = 2$

The two independent upper components then cancel.

Bob moves in one of the middle blocks, $a = 6$ or $a = 9$. The top block $a = 15$ disappears. If Bob chose inner 0, Alice plays $(4, 15)$, leaving the diamond branch at value 1, which cancels the untouched middle block. Otherwise Bob has left a finite normal Apéry board with value $b \geq 3$. Alice plays in that same finite block to leave value 2, which exists because $2 < b$. The three remaining independent upper values are then 2, 1, 3, whose xor is 0.

Bob moves in the top block $a = 15$. If he chose inner 0, Alice removes the whole diamond by playing $(4, 0)$; the two untouched middle blocks cancel. Otherwise the finite top residual has normal value at least 3. Alice plays there to leave value 1. The excluded-ordinal formula restores the three-block value to 3, again canceling the diamond.

Every described reply leaves a position of value 0. The finite moves to a requested smaller number are obtained by the same exact finite-poset recursion used in the verifier. This response analysis is not a claim of a uniform polynomial-time strategy for arbitrary later ordinal-game positions; the winning-strategy existence already follows from the proved zero-value residual and termination.

8 Interpretation, scope, and further questions

8.1 Why the winner changes

The poisoned base game loses because all its first-move residuals have a maximum. In the square, that maximum becomes an entire upper copy of the semigroup. Its interaction with two incomparable lower copies detects a *gap in the option-value spectrum*. The lower pair has value 0, but its first two excluded option values are 0 and 3, not 0 and 2. Putting a value-1 block above it selects the second of these exclusions. The independent Apéry diamond contributes another 3, so the two cancel.

Thus the additional ordinal coordinate is not merely adding more moves to the same strategic situation. It packages the base game into an ordered compound that can observe more than its winner, or even its single Grundy value. This is exactly why a winner-preservation conjecture can fail.

8.2 What the certificate does and does not say

The established universal assertion is a low-value exclusion:

$$g(\text{Ap}(\langle 4, 6, 9 \rangle, t) \setminus \{0\}) \notin \{0, 1\} \quad (t > 0).$$

The raw Grundy values are not claimed to be periodic. In particular, the label 2 in the last column does not mean that every Apéry board has value 2; the early exact values 2, 4, 3, 6, 5 already show otherwise.

Nor is the exponent 2 obtained by testing finitely many exponents. It is established by proving a losing position at exponent 1, giving an explicit winning move at exponent 2, and proving that this same move removes every additional higher-order element.

8.3 Questions left open by this counterexample

A natural next classification problem is to determine which numerical semigroups have first winning exponent 2 and which preserve a second-player win at every ordinal exponent. The diamond-atom criterion supplies a searchable sufficient condition but is not a characterization. Another question is whether the transition exponents of finite natural-number generating sets are bounded by a small finite ordinal. Nothing here establishes such a bound or constructs an exponent above 2.

On the finite-state side, one can ask for a conceptual proof of the specific excluded value that replaces the certificate, or for invariants that compress its 187-entry state. The present proof does not need such a compression: all entries and the transition law are already finite and explicit. The general bounded-Grundy theorem also suggests systematic searches for other small option-spectrum patterns that trigger an ordinal winner change.

9 Reproducing the result

The archive is self-contained apart from a Python interpreter and a LaTeX installation. The mathematical verification uses only Python’s standard library and is written for Python 3.9 or later.

From the extracted directory, run:

```
python code/make_certificate.py
python code/verify_certificate.py
python code/test_certificate.py
```

The generator reproduces `data/certificate.json`. The independent verifier writes `data/verification.json`, `data/exact_boundary_values.csv`, and `data/apery_values.csv`. The last two files contain full integer Grundy values, not only capped labels. The tests also reject deliberately corrupted labels and transition data.

The PDF is built from this source and `certificate_tables.tex`:

```
pdflatex -interaction=nonstopmode -halt-on-error ordinal_chomp_counterexample.tex
pdflatex -interaction=nonstopmode -halt-on-error ordinal_chomp_counterexample.tex
```

The included shell and PowerShell build scripts perform verification before compilation. The code does not need to contact a network service.

A The complete finite-state certificate

Each word has 17 digits in the column order of table 1. Its final digit is the full-tail label. The first eleven words are the initial conditions; every subsequent word is obtained from equation (6.3). The two blocks 226–236 and 254–264 are identical. The table is printed in three independent index–word columns, read downward within each column.

Table 2: All 253 certified bounded-Grundy row vectors.

x	w_x	x	w_x	x	w_x
12	12222122222211102	97	10222122220222122	182	21212002212220222
13	21022222021222202	98	22222001221120222	183	22221222222222212
14	22222112222221202	99	22222222222222222	184	02222021212200022
15	22222222222222222	100	02222022222200022	185	10222222220222122
16	12222122222211122	101	20222222220222222	186	22222001222220222
17	21222222212222222	102	12221001221120222	187	22222222222222222
18	22222112222221202	103	21122222212222222	188	02222022122200012
19	22222222222222222	104	02112022222200012	189	20222222220222222
20	12222122220211102	105	20221222220212222	190	22222002222220222
21	21222222212222222	106	12221001221120222	191	22222222222222222
22	22222112222221202	107	21122222212222222	192	02221021222200022
23	22222222222222222	108	02122022222200012	193	20222222220222222
24	12222122222211122	109	20222221220222222	194	22222002222220222
25	21222222212222222	110	12222001221220222	195	22122222122222212
26	22202112222221202	111	21222222212222222	196	02212022122200012
27	22220222222202222	112	02222022222200022	197	20221222220212222
28	12222122220211102	113	20221221220222222	198	21222002111220222
29	21222222212222222	114	22222002221220222	199	22222222222222222
30	22222112222221202	115	22222222222222222	200	02221021222200022
31	22222222222222222	116	02222021222200022	201	20222222220222222
32	12222102222211102	117	20122222220222222	202	22212002212220222
33	21222222212222222	118	22221002222210222	203	22222222122222212
34	22222112222221222	119	22222222222222222	204	02222022222200022
35	22022222022222202	120	01112022122200012	205	20222222220222222
36	12222122222211102	121	20221222220212222	206	22222002222220222
37	21222222212222222	122	22212002122220212	207	22222222222222222
38	22222112220221202	123	12221122222212122	208	02122022122200012
39	22222222222222222	124	02221022222200012	209	10222221210222222
40	12222122222211102	125	20222222220222222	210	21122002221220222
41	21222222212222222	126	22222001212220222	211	22212222122222212
42	22222102212221202	127	22222222222222222	212	02222022222200022
43	22222222222222222	128	02222022222200022	213	20221222220212222
44	12222122222211102	129	20222222220222222	214	22222002222220222
45	21220222221202222	130	22222002222220222	215	21212222212221222
46	22222112222221222	131	22222222222222222	216	02222022122200012
47	22222222222222222	132	02222022121100012	217	20222221220222222
48	12222122222211102	133	20222222220222222	218	21222002221220222
49	21222222212222222	134	21222002221220212	219	22222222222222222
50	22222112220221202	135	22222222222222222	220	02222022222200022
51	22222222222222222	136	01212022212200022	221	20221222220212222
52	12222122222211122	137	20221222220212222	222	22222002222220222
53	21222222212222222	138	22212002212220222	223	12221121222222122
54	22222102212221202	139	12222122222222122	224	02222021222200022
55	22222222222222222	140	02222021222200022	225	20122222220222222
56	12222122220211102	141	20222222220222222	226	22221002222210222
57	21022222021222202	142	22222002121120212	227	22222222222222222
58	22222112222221202	143	22222222222222222	228	02222022222200022
59	22222222222222222	144	02222022222200022	229	20122222120222212
60	12222102222211102	145	20222222220222222	230	12222001222220222
61	21222222212222222	146	21212002212220222	231	21122222212222222
62	22222112222221202	147	22221222222212222	232	02221022222200012
63	22022222222202222	148	02212012222200012	233	20222222220222222
64	12222122222211102	149	10222122220222122	234	22122002122220212

Continued on the next page

x	w_x	x	w_x	x	w_x
65	21222222221222222	150	22222002122220212	235	22122222222212222
66	22222112222221202	151	22122222122222222	236	02112022222200012
67	22222222222222222	152	02221022222200012	237	20212222120222212
68	12222122222211122	153	20222222220222222	238	22222002212220222
69	21222222221222222	154	22222002222220222	239	22222222122222212
70	22222112222221222	155	22221221222222222	240	02222022222200022
71	22222222222222222	156	02222021222200022	241	20222222220222222
72	122221222220211102	157	20122222220222222	242	22222002222220222
73	21222222221222222	158	22222002222210222	243	22222222222222222
74	22222112220221202	159	22222222222222222	244	022220222122200012
75	22222222222222222	160	02222022121100012	245	20222222220222222
76	10222122202211022	161	20222222220222222	246	22221002222210222
77	21222222221222222	162	22222002221220212	247	22222222222222222
78	22202102212221122	163	22222222222222222	248	02222022222200022
79	02222022222222022	164	01222022212200022	249	20221222220212222
80	12222122220221202	165	20222222220222222	250	22222002222220222
81	21022222202122222	166	22212002212220222	251	12221121222222122
82	22222112222201222	167	12222122222222122	252	02222022222200022
83	22222222222222222	168	02222021221200022	253	20222222220222222
84	10222122202210022	169	20222222220222222	254	22221002222210222
85	21222222221222222	170	22222002221220212	255	22222222222222222
86	12222120212221122	171	22222222222222222	256	02222022222200022
87	01222022221222022	172	01222022211200022	257	20122222120222212
88	12222021222100022	173	20222222220222222	258	12222001222220222
89	21222222221222222	174	21212002212220222	259	21122222221222222
90	21222002121220212	175	12222122222222122	260	02221022222200012
91	22222222222222222	176	02222011222200022	261	20222222220222222
92	01212022212200022	177	10222122220222222	262	22122002122220212
93	10221122220212122	178	22222002121120212	263	22122222222212222
94	21222001212220122	179	22122222122222212	264	02112022222200012
95	22222222222222222	180	02221022222200012		
96	02222021212200022	181	20222222220222222		

B A compact statement of the checking algorithm

The mathematical core of the recurrence check can be written as follows. Here `tails` is the list of gap ideals, `index` maps each ideal to its column, rows contains already checked words, and `member(n)` tests $n \geq 0$ and $n \notin \{1, 2, 3, 5, 7, 11\}$. The actual verifier independently reconstructs that membership test from the generators.

```
def next_row(x, rows, tails, index):
    result = []
    for C in tails:          # increasing cardinality
        options = set()
        for d in range(1, 12):
            T = frozenset(g for g in gaps
                           if g < d or g - d in C)
            options.add(rows[x - d][index[T]])
        for a in C:
            R = frozenset(g for g in C if not member(g - a))
            options.add(result[index[R]])
        value = 0
        while value in options:
            value += 1
        result.append(min(2, value))
    return tuple(result)
```

The omitted far-prefix contribution is justified by checking that the initial exceptional Apéry values and all full-tail coordinates are at least 2. The complete state is then an eleven-row window. Once that window repeats, the same deterministic operation repeats forever.

For the finite seed calculation, and for the independent exhaustive check, the only game-theoretic recursion is

$$g(A) = \text{mex} \{g(\{z \in A : z - y \notin \mathcal{S}\}) : y \in A\}, \quad g(\emptyset) = 0. \quad (\text{B.1})$$

All sets A in this equation are finite sets of positive integers. Each recursive argument is a proper subset, so the calculation terminates by induction on $|A|$. The bit-mask verifier implements this very recurrence with exact, unrestricted nonnegative-integer values.

References

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- [4] Fabián Rivero Herrera, *Hanf numbers for poset games*, arXiv:2504.07317v3, 5 January 2026. See Example 3.2 and Question 4.6.